UNIVERSITY OF THE PUNJAB

Third Semester 2015
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Mathematics A-III
Course Code: MATH-201

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Short Questions (2x10=20)

Q2.

i. Define a "Nilpotent matrix"

ii. Define a "Hermition matrix"

iii. Let $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$, $|A| \neq 0, A \neq I$, then find a ,b and c if A is involutary.

iv. If A is a 3x3 matrix with detA = 2 , then find det A^5.

v. Prove that $\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & a & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

vi. Show that the set {(1,3),(2,5)} is linearly independent but the set {(1,3),(2,6)} is linearly dependent in R^2.

vii. Check whether W= {(x,y,z) : x+y+z=0} is a subspace of R^3.

viii. Find the dimension of subspace R^4, spanned by {(-1,-1,5,0),(0,0,0,1)}.

ix. Find the eigen values for the linear transformation $T:R^2 \rightarrow R^2$, such that

$$T(x,y) = (3x+3y, x+5y)$$

x. Show that A and A^t have same eigen values, where A is a square matrx.

Subjective Questions (6x5=30)

Q3. Use Gauss-Jordan method to reduce the given system to reduced echelon form and determine solution.

$$6x - 6y + 6z = 6$$
$$2x - 4y - 6z = 12$$
$$10x - 5y - 5z = 30$$

Q4. If A and B are 3x3 matrices such that $\det(A^2B^2) = 108$ and $\det(A^2B^3) = 72$, find det(2A) and $\det(B^{-1})$.

Q5. Determine 'k' so that the vectors (1,-1,k-1) , (2,k,-4), (0,2+k,-8) in R^3 are linearly independent.

Q6. Let U and W be 2-dimensional subspaces of R^3, show that $U \cap W = \{0\}$.

Q7. Show that the linear transformation $T:R^2 \rightarrow R^2$ defined by

$$T(x,y) = (x+y, x-y, x+2y)$$

Is one-one.

Q8. Find an orthogonal matrix whose first row is $\left(0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$.

UNIVERSITY OF THE PUNJAB

Roll No. ....

Third Semester 2015

Examination: B.S. 4 Years Programme

PAPER: Mathematics A-III
Course Code: MATH-201

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE

Multiple choice questions. **(1x10=10)**

Q1. **Encircle the correct answer.**

i) A square matrix A is said to be periodic if

a) $A^2=A$ b) $A^{k+1}= A$

c) $A^k = 0$ d) $A^k =A$

ii) The number of non-zero rows in echelon form of a matrix A determine

a) The rank of A b) the period of A

c) A is nilpotent d) A is idempotent

iii) Matrix A is involu-tary , if $A^2=$

a) I b) 0

c) A d) none of these

iv) If A is an nxn matrix and λ is a scalar, than $\det(\lambda A) =$

a) 0 b) 1

c) $\lambda^n det A$ d) $\lambda\ det A$

v) The system of linear equations AX=B with m=n has a unique solution, if A is

a) singular b) non-singular

c) periodic d) equal to B

vi) The empty set Φ of a vector space V(F) is always taken as

a) linearly independent b) linearly dependent

c) basis d) subspace

vii) Let U be a non-empty subset of a vector space V(F), then U is said to be a basis of V(F) if

a) U is linearly independent subset of V(F) b) U spans V(F)

c) U is linearly dependent subset of V(F) d) both a and b

viii) The number m of vectors in the basis of a vector space V(F) is called the

a) Dimension of V(F) b) subspace

b) spanning set d) dimension of m

ix) A linear transformation that is both one-one and onto is called

a) Homomorphism b) isomorphism

c) Bijective d) none of these

x) Non-zero eigen vectors of a matrix A corresponding to distinct eigen values are

a) linearly independent b) linearly dependent

c) orthogonal d) none of these

UNIVERSITY OF THE PUNJAB

Roll No. ....

Third Semester 2017

Examination: B.S. 4 Years Programme

PAPER: Mathematics A-III | **TIME ALLOWED: 30 mins.**

Course Code: MATH-201/MTH-21309 | **MAX. MARKS: 10**

Attempt this Paper on this Question Sheet only.

OBJECTIVE

Multiple Choice Questions (1x10=10)

Q1. Encircle the correct answer.

i) A square matrix A is said to be symmetric, if:
a) $A^T = A^{-1}$ b) $A^T = A$ c) $A^T = 0$ d) $A^T = -A$

ii) The empty set of a vector space V(F) is always taken as
a) Linearly independent b) linearly dependent c) basis d) subspace

iii) Which of the following set is a vector space?
a) Set of real numbers b) Set of natural numbers c) Set of whole numbers d) Set of integers

iv) For a nonsingular matrix A, $(A^n)^{-1}$ = ...
a) $(A^{-1})^n$ b) A^{-1} c) A^n d) $(A^{-1})^T$
Where, n is a positive integer.

v) A system of linear equations $Ax=b$, with m=n has a unique solution, if A is
a) Singular b) Non singular c) Periodic d) Idempotent

vi) For the direct sum of two sub spaces U and W of a vector space V,
then $U \cap W$ = ...
a) $\{0\}$ b) U c) W d) empty set

vii) A linear transformation that is both one-one and onto is called:
a) Bijective b) Homomorphism c) Isomorphism d) Basis

viii) A set of linearly independent vectors spanning a vector space V is called ....
a) Basis for V b) dimension of V c) column space d) row space

ix) The eigen values of a symmetric matrix are:
a) Orthogonal b) Real and equal c) Complex d) Real

x) A linear transformation $T: U \to V$ is one-one if and only if
a) R(T)= $\{0\}$ b) N(T) = $\{0\}$ c) R(T)=N(T) d) R(T)=V

UNIVERSITY OF THE PUNJAB

Third Semester 2017
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Mathematics A-III
Course Code: MATH-201/MTH-21309

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q2. Short Questions (2x10=20)

i) Let A and B be 3x3 matrices with $det(A)=2$ and $det(B)=4$. Find the values of $det(AB)$ and $det(A^{-1}B)$.

ii) Consider a linear system whose augmented matrix is of the form

$$\begin{bmatrix} 1 & 2 & 1 & 1 \\ -1 & 4 & 3 & 2 \\ 2 & -2 & a & 3 \end{bmatrix}$$ For what value(s) of a will the system have a unique solution?

iii) Define Undetermined system

iv) Define determinant of a matrix.

v) Check whether the transformation $T: R^2 \to R^2$, defined as $T(x,y)=(-y,x)$ is linear or not.

vi) Check whether $W=\{(x,0,z): x,y,z \in R\}$ is a subspace of R^3?

vii) Find the eigen values of linear transformation $T: R^2 \to R^2$, defined as $T(x,y)=(3x+3y, x+5y)$.

viii) Without expansion, show that

$$\begin{vmatrix} a+b & c & 1 \\ b+c & a & 1 \\ c+a & b & 1 \end{vmatrix} = 0$$

ix) Show that if A is a symmetric nonsingular matrix, then A^{-1} is also symmetric.

x) Let $A=\begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$

Compute A^2 and A^3. What will A^n turn out to be?

<u>Subjective Questions (5x6=30)</u>

Q3. Determine the solution of the following system of linear equations

$$x-y+z=1$$
$$x-2y-3z=6$$
$$2x-y-z=6$$

Q4. Find equations defining the subspace W of R^3 spanned by the vector (2,3,4).

Q5. Check whether the following transformation $T: R^3 \to R^2$ is one-one or not $T(x,y,z)=(x-y,z)$.

Q6. Let $T: R^3 \to R^3$ be a linear transformation defined by $T(a,b,c)=(a+2b-c,\ b+c,\ a+b-2c)$. Find basis and dimension of R (T) and N (T).

Q7. Prove that a one-one linear transformation preserves basis and dimension.

Q8. If A is a matrix over R and $AA^T=0$, then show that A=0.

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Third Semester 2018
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PAPER: Mathematics A-III | **TIME ALLOWED: 2 hrs. & 30 mins.**
Course Code: MATH-201/MTH-21309 | **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE
SECTION – II

Q. 2	SHORT QUESTIONS	
(i)	For the given matrix find the basis of Column space. $$\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix}$$	(4)
(ii)	If A is a nonsingular matrix whose inverse is $\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, find A.	(4)
(iii)	Find the reduced echelon form of the matrix $\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$	(4)
(iv)	Prove that $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$	(4)
(v)	Check whether W is a subspace of V or not? $V = R^3,\ W = \{(a,b,c) \in V:\ a^2 + b^2 + c^2 \le 1\}$.	(4)

SECTION – III

LONG QUESTIONS		
Q.3	If possible, find the inverse of the matrix $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$	(6)
Q.4	Determine whether the vectors are linearly independent or not? $v_1 = (1,-2,3), v_2 = (5,6,-1), v_3 = (3,2,1)$.	(6)
Q.5	Determine whether or not the set of vectors $\{(1,2,-1),(0,3,1),(1,-5,3)\}$ is a basis for R^3	(6)
Q.6	Find the real orthogonal matrix P for which $P^{-1}AP$ is orthogonal where $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$	(6)
Q.7	Determine the values of a for which the system of linear equations has no solution, exactly one solution and infinitely many solutions. $x + y + 7z = -7$ $2x + 3y + 17z = -16$ $x + 2y + (a^2+1)z = 3a$.	(6)

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Third Semester 2018
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PAPER: Mathematics A-III | **TIME ALLOWED: 30 mins.**
Course Code: MATH-201/MTH-21309 | **MAX. MARKS: 10**

Attempt this Paper on this Question Sheet only.

OBJECTIVE

SECTION – I

Q.1	MCQs (1 mark each)
(i)	If $\left(\overline{A}\right)' = -A$ then A is called **(a)** Symmetric matrix **(b)** Skew symmetric matrix **(c)** Hermitian matrix **(d)** Skew Hermitian matrix
(ii)	The dimension of $KerT$ is called .... **(a)** Rank **(b)** Nullity **(c)** Column space **(d)** None of these
(iii)	A system of m homogeneous linear equations $Ax = 0$ in n variables has a non-trivial solution if and only if the rank of A is --------- **(a)** equal to n **(b)** less than n **(c)**greater than n **(d)** none of these
(iv)	A unit vector orthogonal to both $(1, 1, 2)$ and $(0, 1, 3)$ in R^3 is --------- **(a)** $\left(\frac{1}{\sqrt{11}}, \frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}\right)$ **(b)** $\left(\frac{-1}{\sqrt{11}}, \frac{3}{\sqrt{11}}, \frac{1}{\sqrt{11}}\right)$ **(c)** $\left(\frac{2}{\sqrt{11}}, \frac{-3}{\sqrt{11}}, \frac{-1}{\sqrt{11}}\right)$ **(d)** $\left(\frac{-1}{\sqrt{11}}, \frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}\right)$
(v)	The characteristic polynomial of the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ is............... **(a)** $p(\lambda) = (2-\lambda)^2$ **(b)** $p(\lambda) = (2-\lambda)(3-\lambda)$ **(c)** $p(\lambda) = \lambda^2$ **(d)** None of these
(vi)	The property $\forall$ a, b$\in$R then $a + b \in R$ is called **(a)** Associative property **(b)** Transitive property **(c)** Closure property **(d)** None of these
(vii)	The subspace of R^3 spanned by the vector (a, b, c) is --------- **(a)** $x = t, y = bt, z = ct$ **(b)** $x = -at, y = -bt, z = -ct$ **(c)** $x = at, y = bt, z = ct$ **(d)** None of these
(viii)	Let R^3 be the vector space of all ordered triples of real numbers. Then the transformation $T : R^3 \to R^3$ defined by $T(x, y, z) = (x, y, 0)$ is **(a)** Linear **(b)** Not Linear **(c)** Rational **(d)** None of these
(ix)	A linear transformation that is both one-one and onto is called **(a)** Isomorphism **(b)** Homomorphism **(c)** Basis **(d)** Bijective
(x)	A linear transformation $T : U \to V$ is one-to-one if and only if --------- **(a)** $N(T) = \{0\}$ **(b)** $N(T) \neq \{0\}$ **(c)** $N(T) = \{1\}$ **(d)** $N(T) = \{-1\}$

UNIVERSITY OF THE PUNJAB

Third Semester – 2019

Examination: B.S. 4 Years Program

Roll No. in Fig. ....

Roll No. in Words. ....

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PAPER: Mathematics A-III

Course Code: MATH-201/MTH-21309 Part–I(Compulsory)

MAX. TIME: 30 Min.

MAX. MARKS: 10

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Signature of Supdt.:

Attempt this Paper on this Question Sheet only.

Please encircle the correct option. Division of marks is given in front of each question.

This Paper will be collected back after expiry of time limit mentioned above.

Q.1. Encircle the right answer, cutting and overwriting is not allowed. (1x10=10)

(i)	If $(\overline{A})' = -A$ then A is called (a) Symmetric matrix (b) Skew symmetric matrix (c) Hermitian matrix (d) Skew Hermitian matrix
(ii)	If W is a linear subspace of V then (a) $\dim(W) \le \dim(V)$ (b) $\dim(W) \ge \dim(V)$ (c) $\dim(W) = \dim(V)$ (d) None of these
(iii)	A set of linear equations is represented by the matrix equation Ax = b. The necessary condition for the existence of a solution for this system is (a) A must be invertible (b) b must be linearly depended on the columns of A (c) b must be linearly independent on the columns of A (d) none of these
(iv)	A unit vector orthogonal to both $(1, 1, 2)$ and $(0, 1, 3)$ in R^3 is ---------- (a) $\left(\frac{1}{\sqrt{11}}, \frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}\right)$ (b) $\left(\frac{-1}{\sqrt{11}}, \frac{3}{\sqrt{11}}, \frac{1}{\sqrt{11}}\right)$ (c) $\left(\frac{2}{\sqrt{11}}, \frac{-3}{\sqrt{11}}, \frac{-1}{\sqrt{11}}\right)$ (d) $\left(\frac{-1}{\sqrt{11}}, \frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}\right)$
(v)	The characteristic polynomial of the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ is............... (a) $\dim(W) = \dim(V)$ (b) $p(\lambda) = (2-\lambda)(3-\lambda)$ (c) $p(\lambda) = 0$ (d) None of these
(vi)	The property $\forall$ a, b$\in$R then $a+b=b+a=b \in R$ then a is called (a) Identity (b) Inverse (c) Conjugate (d) None of these
(vii)	The subspace of R^3 spanned by the vector (a, b, c) is --------- (a) $x=t, y=bt, z=ct$ (b) $x=-at, y=-bt, z=-ct$ (c) $x=at, y=bt, z=ct$ (d) None of these
(viii)	Let R^3 be the vector space of all ordered triples of real numbers. Then the transformation $T: R^3 \to R^3$ defined by T(x,y,z)=(x,y,z) is (a) Identity (b) Not Linear (c) Rational (d) None of these
(ix)	A linear transformation T that is both one-one and onto is has Ker(T) (a) 0 (b) 1 (c) 2 (d) None of these
(x)	Let V be vector space of finite dimension n. Then any $n+1$ or more vectors in V is --------- (a) linearly dependent (b) linearly independent (c) both (a) and (b) (d) none of these.

UNIVERSITY OF THE PUNJAB

Third Semester – 2019

Examination: B.S. 4 Years Program

Roll No. ………………..

PAPER: Mathematics A-III **MAX. TIME: 2 Hrs. 30 Min.**

Course Code: MATH-201/MTH-21309 Part – II **MAX. MARKS: 50**

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Questions with Short Answers. **(5x4=20)**

(i)	Decide yes or no about the following statement, if no justify by counter example. If A is 2×2 matrix such that $A^2 = \mathbf{0}$ Then A must be a null matrix-	(4)
(ii)	If possible express $\vec{v} = (2,-5,3)$ in R^3 as a linear combination of the vectors $\vec{u}_1 = (1,-3,2)$, $\vec{u}_2 = (2,-4-1)$, $\vec{u}_3 = (1,-5,7)$.	(4)
(iii)	Check whether W is a subspace of V or not. $V = \{f : \ f : R \to R\}$, $W = \{f \in V : \ f(1) = 0\}$.	(4)
(iv)	Show that $\det \begin{bmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{bmatrix} = (a-1)^3(a+3)$	(4)
(v)	If A and B are invertible matrices of same order and AB is invertible then show that $(AB)^{-1} = B^{-1}A^{-1}$.	(4)

P.T.O.

LONG QUESTIONS (5x6=30)		
Q.3	Solve the following system of linear equations by using an appropriate method. $$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 3 \\ 2x_1 + x_2 - x_3 &= 9 \\ -3x_1 - 4x_2 + 5x_3 &= -15. \end{aligned}$$	(6)
Q.4	Define $T: R^3 \to R^3$ by $T(x_1, x_2, x_3) = (-x_3, x_1, x_1 + x_3)$. Find $N(T)$. Is T one-to-one?	(6)
Q.5	If $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ determine the value of $A^2 - 4A - 5I$.	(6)
Q.6	Find the real orthogonal matrix P for which $P^{-1}AP$ is orthogonal where $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$	(6)
Q.7	Express matrix M as a linear combination of the matrices A, B and C where $M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$.	(6)

UNIVERSITY OF THE PUNJAB

Third Semester – 2019

Examination: B.S. 4 Years Program

(Clash)

Roll No. in Fig. ....

Roll No. in Words. ....

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PAPER: Mathematics A-III

Course Code: MATH-201/MTH-21309 Part–I(Compulsory)

MAX. TIME: 30 Min.

MAX. MARKS: 10

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Signature of Supdt.:

Attempt this Paper on this Question Sheet only.
Please encircle the correct option. Division of marks is given in front of each question.
This Paper will be collected back after expiry of time limit mentioned above.

Q.1. Encircle the right answer, cutting and overwriting is not allowed. (1x10=10)

(i)	Rank of A is --------- Rank of A^T (a) equal (b) greater than (c) less than (d) None of these
(ii)	If A and B are *symmetric* matrices. Then AB is also (a) symmetric (b) skew-symmetric (c) Hermitian (d) None of these
(iii)	If A is a matrix of order 3×3 and $\det(A)=-1$, then the value of $\det(2A)$ is -------- (a) -8 (b) -6 (c) 8 (d) 18
(iv)	The dimension of $KerT$ is called .... (a) Rank (b) Nullity (c) basis (d) none of these
(v)	The characteristic polynomial of the matrix $\begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$ is................ (a) $p(\lambda)=(1-\lambda)^2$ (b) $p(\lambda)=(5-\lambda)(1-\lambda)$ (c) $p(\lambda)=0$ (d) None of these
(vi)	If $(\overline{A})' = -A$ then A is called.................matrix (a) Symmetric (b) Skew symmetric (c) Hermitian (d) Skew Hermitian
(vii)	The property $\forall$ a, b $\in$R then $(a+b)=(b+a)$ is called (a) Closure property (b) Transitive property (c) Commutative property (d) Associative property
(viii)	A linear transformation $T:U\to V$ is one-to-one if and only if ---------- (a) $N(T)=\{0\}$ (b) $N(T)\neq\{0\}$ (c) $N(T)=\{1\}$ (d) $N(T)=\{-1\}$
(ix)	Let R^3 be the vector space of all ordered triples of real numbers. Then the transformation $T:R^3\to R^3$ defined by $T(x,y,z)=(x,y,0)$ is (a) Linear (b) Not Linear (c) Rational (d) None of these
(x)	A symmetric matrix of order n has -------- eigen values. (a) n-1 (b) n+1 (c) n (d) 0

UNIVERSITY OF THE PUNJAB

Third Semester – 2019

Examination: B.S. 4 Years Program

(Clash)

Roll No. ....

PAPER: Mathematics A-III

Course Code: MATH-201/MTH-21309 Part – II

MAX. TIME: 2 Hrs. 30 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q. 2	SHORT QUESTIONS	
(i)	Show that the functions $f(t)=\cos t$, $g(t)=\sin t$, $h(t)=t$ from R to R are linearly independent.	(4)
(ii)	Without evaluating, verify that $\det\begin{bmatrix} b+c & b & c \\ c & c+a & a \\ b & a & a+b \end{bmatrix} = 2a(b^2+c^2)$.	(4)
(iii)	Show that the matrix $\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ is involutery.	(4)
(iv)	Check whether W is a subspace of V or not. $V=R^2$ and W is the set of first quadrant vectors of R^2.	(4)
(v)	For the given matrix find the basis of Column space. $\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 0 \end{bmatrix}$	(4)

P.T.O.

	LONG QUESTIONS	
Q.3	Solve the system of linear equations by Gaussian elimination method $x+y+2z=9, \qquad 2x+4y-3z=1, \qquad 3x+6y-5z=0$	(6)
Q.4	If possible, find the inverse of the matrix $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$	(6)
Q.5	For a real number k, consider the following system. For what value k does the system have a solution. $\begin{aligned} x_1+x_2+kx_3+x_4 &= k \\ -x_2+x_3+2x_4 &= 0 \\ x_1+2x_2+x_3-x_4 &= -k. \end{aligned}$	(6)
Q.6	Determine whether the vectors are linearly independent or not? $v_1=(1,-2,3), v_2=(5,6,-1), v_3=(3,2,1)$.	(6)
Q.7	If possible find a matrix P that diagonalize the following matrix. $A=\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	(6)